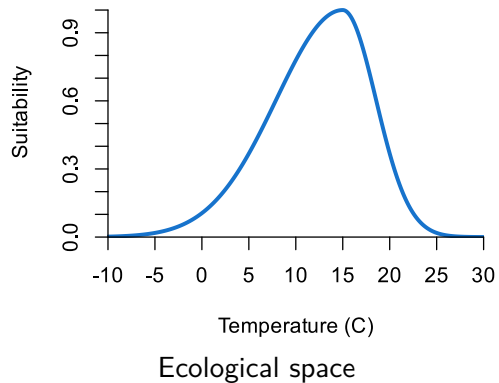


Fitting ENMs

Generalized Additive Models (GAMs)

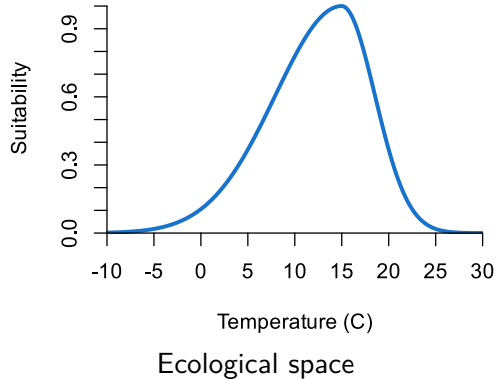
Duality of SDMs

Ecological niche model (ENM)

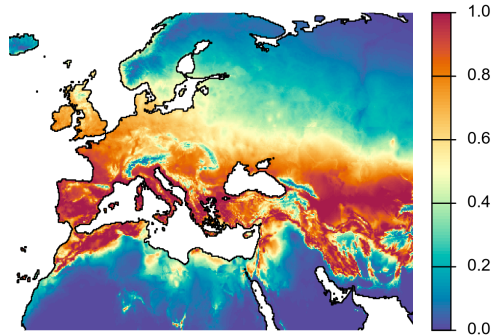


Duality of SDMs

Ecological niche model (ENM)



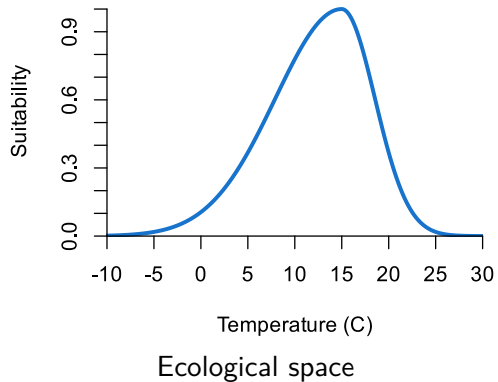
Species distribution model (SDM)



Geographic space

Ecological Niche Model (ENM)

Ecological niche model (ENM)



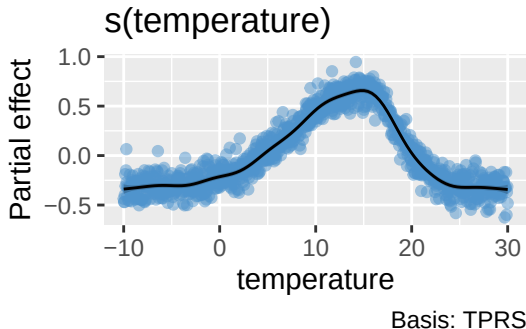
Introduction to GAMs

Generalized additive models (GAMs)

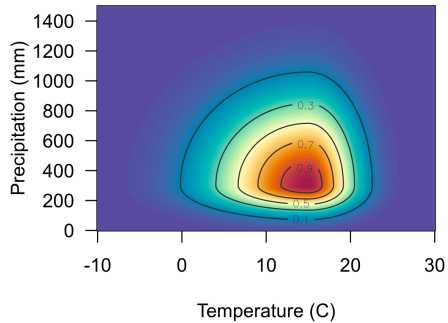
- ▶ Machine learning statistical framework
- ▶ Highly flexible and extremely powerful
- ▶ Easy to interpret
- ▶ Easy to implement in R (`mgcv` & `gratia`)

Example – Fitting a 1D niche

```
library(mgcv)
library(gratia)
d <- read.csv("../data/niche-1d.csv")
enm <- gam(suitability ~ s(temperature, k = 20), data = d)
draw(enm, residuals = TRUE, rug = FALSE)
```

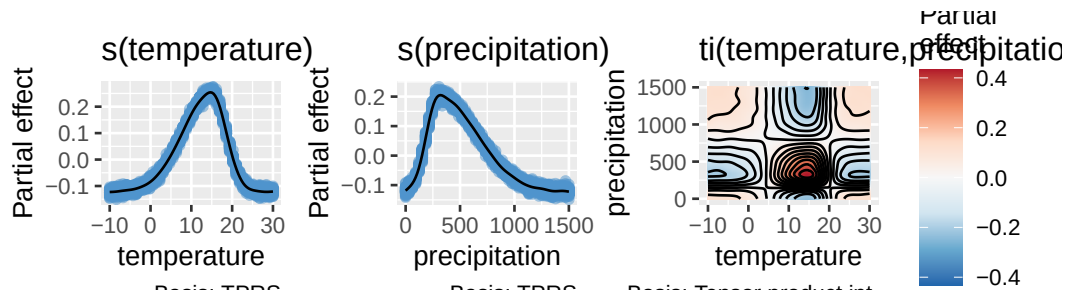


Example – Fitting a 2D niche

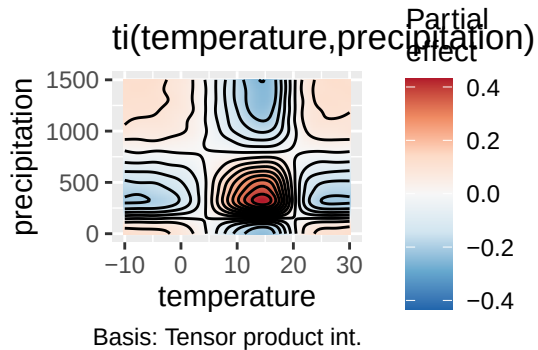


Example – Fitting a 2D niche

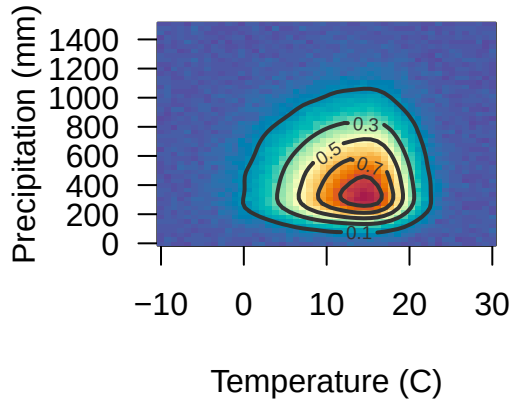
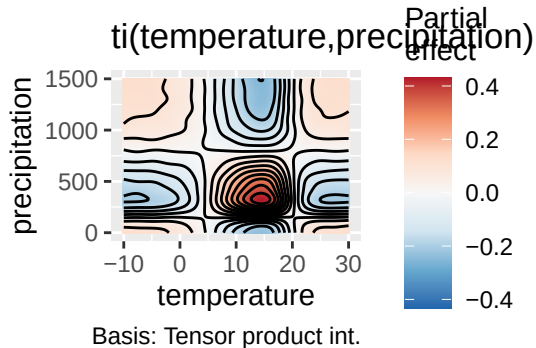
```
d <- read.csv("../data/niche-2d.csv")
enm <- gam(
  suitability ~ s(temperature, k = 20) + s(precipitation, k = 20) +
    ti(temperature, precipitation, k = 16),
  data = d
)
draw(enm, residuals = TRUE, rug = FALSE, nc = 3)
```



Example – Fitting a 2D niche



Example – Fitting a 2D niche



GAM workflow

1. Fit GAM model: `gam(<y> ~ <x>)`

GAM workflow

1. Fit GAM model: `gam(<y> ~ <x>)`
2. Simple covariate: `s(<x>, k = ...)`

GAM workflow

1. Fit GAM model: `gam(<y> ~ <x>)`
2. Simple covariate: `s(<x>, k = ...)`
3. Interaction term: `ti(x1, x2, k = ...)`

GAM workflow

1. Fit GAM model: `gam(<y> ~ <x>)`
2. Simple covariate: `s(<x>, k = ...)`
3. Interaction term: `ti(x1, x2, k = ...)`
4. Inspect: `draw(...)`

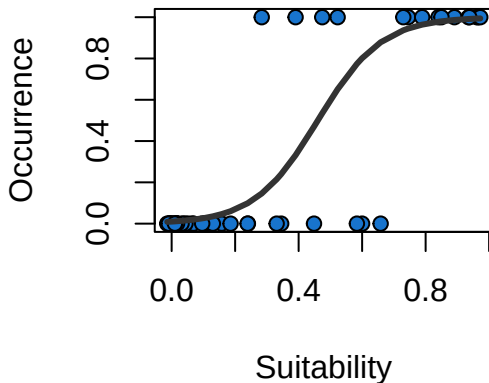
GAM workflow

1. Fit GAM model: `gam(<y> ~ <x>)`
2. Simple covariate: `s(<x>, k = ...)`
3. Interaction term: `ti(x1, x2, k = ...)`
4. Inspect: `draw(...)`
5. Appraise: `k.check(...)`
 - ▶ $k - index \ll 1 \rightarrow \text{Increase } k$
 - ▶ $p - value < 0.05 \rightarrow \text{Increase } k$

Data for ENM/SDM

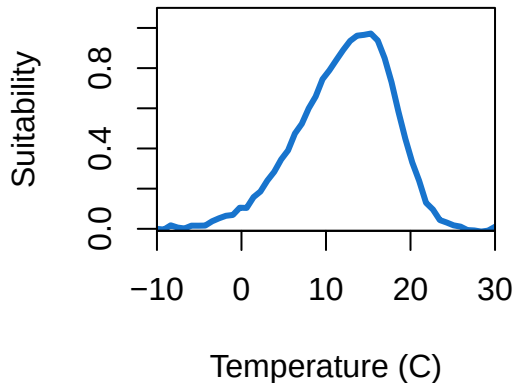
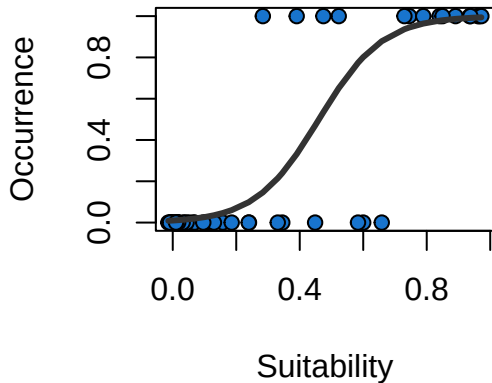
GAM – SDM data

- ▶ Rarely (if ever) you will get suitability as data
- ▶ More likely (almost always) you will get presence (1) and absence (0) data



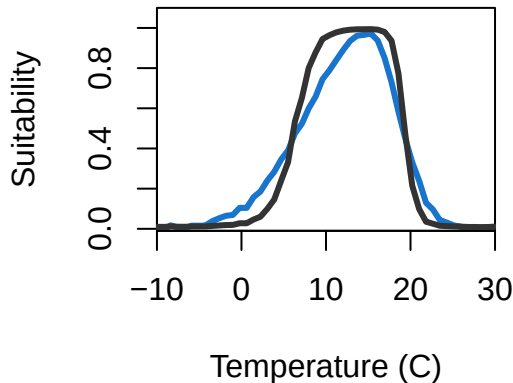
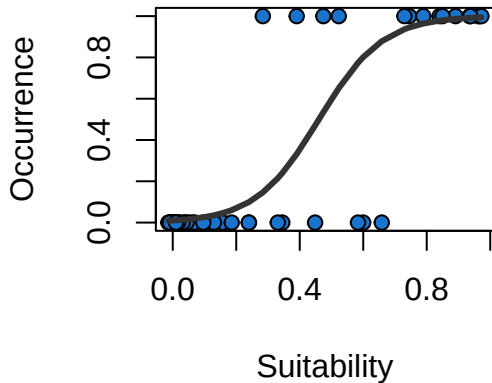
GAM – SDM data

- ▶ Rarely (if ever) you will get suitability as data
- ▶ More likely (almost always) you will get presence (1) and absence (0) data



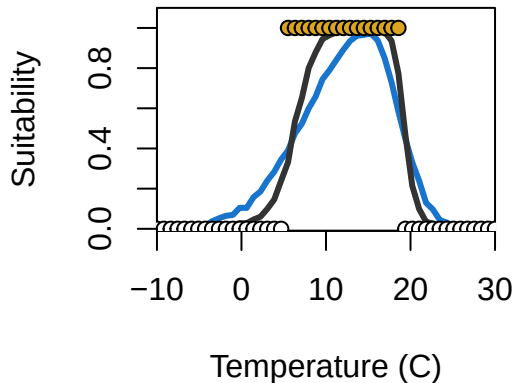
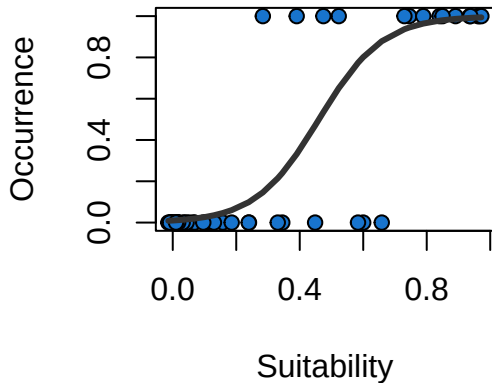
GAM – SDM data

- ▶ Rarely (if ever) you will get suitability as data
- ▶ More likely (almost always) you will get presence (1) and absence (0) data



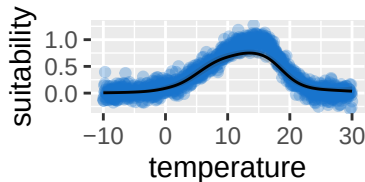
GAM – SDM data

- ▶ Rarely (if ever) you will get suitability as data
- ▶ More likely (almost always) you will get presence (1) and absence (0) data



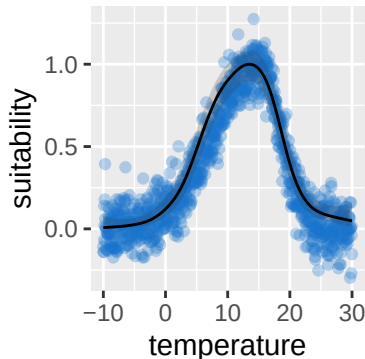
GAM – Example with SDM data

```
d <- read.csv("../data/niche-1d.csv")
enm <- gam(occurrence ~ s(temperature, k = 30), data = d, family = binomial)
# draw(enm, rug = FALSE, residuals = TRUE)
library(ggplot2)
fitted_values(enm) |>
  ggplot() +
  geom_point(data = d, aes(temperature, suitability), alpha = .3, col = "darkblue") +
  geom_ribbon(aes(x = temperature, ymin = .lower_ci, ymax = .upper_ci), alpha = .5, col = "darkblue") +
  geom_line(aes(temperature, .fitted))
```



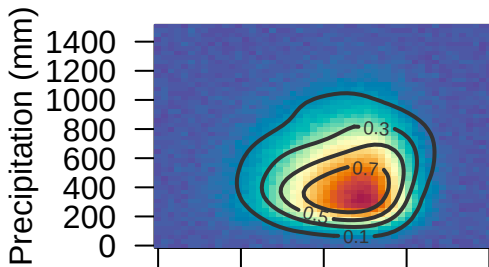
GAM – Fitted niche is scaled

```
fitted_values(enm) |>  
  ggplot() +  
  geom_point(data = d, aes(temperature, suitability), alpha = .3, col = "d  
  geom_ribbon(aes(x = temperature, ymin = .lower_ci / max(.fitted), ymax =  
  geom_line(aes(temperature, .fitted / max(.fitted)))
```



GAM – 2D example with SDM data

```
d <- read.csv("../data/niche-2d.csv")
enm <- gam(
  occurrence ~ s(temperature, k = 10) + s(precipitation, k = 10) +
    ti(temperature, precipitation, k = 10),
  data = d,
  family = binomial
)
niche <- predict(enm, type = "response")
```



Model selection

AIC

You can perform model selection on ENMs just as any other statistical models.

```
enm_simple <- gam(  
  occurrence ~ s(temperature, k = 10) + s(precipitation, k = 10),  
  data = d, family = binomial, method = "ML"  
)  
enm_int <- gam(  
  occurrence ~ s(temperature, k = 10) + s(precipitation, k = 10) +  
    ti(temperature, precipitation, k = 10),  
  data = d, family = binomial, method = "ML"  
)  
AIC(enm_simple, enm_int)
```

	df	AIC
enm_simple	12.66876	1126.237
enm_int	28.21966	1113.738

Maximum Likelihood

When you perform model selection using AIC, remember to specify `gam(..., method = "ML")`.

Once you select the model, you can update it and remove `method = "ML"`.